

Hanif Bin Ariffin

Systems Engineer

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GitHub
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TECHNICAL SKILLS

Languages

C++ | Rust | C | Go | Python | TypeScript | VHDL

Systems & Low Latency

Kernel Bypass (ef_vi) | Solarflare | Lock-free structures | Shared Memory | HFT Infrastructure

Data & Infrastructure

Redis Internals | Market Data | Big Data (TB scale) | Linux/Unix | Docker

EDUCATION

BSc in Computer Engineering
University of Ottawa
Aug 2015 – Aug 2020
Canada

Systems engineer specializing in high-performance computing and low-latency infrastructure, with a proven track record of optimizing performance-critical C++ applications, processing terabytes of market data, and contributing to major open-source projects, including Redis and Rust Coreutils.

WORK EXPERIENCE

Software Developer
Dec 2023 – Present
Confinex Technologies | *Pulau Pinang*

Developing and maintaining a complete infrastructure stack for high-frequency trading.

- Optimized a lock-free C++ shared-memory queue library.
- Developed a market data processing library supporting data formats from multiple exchanges.
- Built kernel-bypass C++ applications using Solarflare's ef_vi API for direct NIC access.
- Engineered high-performance C++ bridges for Python JIT interaction with brokers.
- Automated the processing of terabytes of market data from exchanges worldwide.
- Developed an FPGA-based market data processor for the ExaNIC X25 card.

Senior Full-Stack Developer
Feb 2022 – Jul 2022
Globelise | *Singapore*

Designed and implemented the backend for a global hiring management system.

- Led the development of third-party payroll integrations for international markets.
- Implemented a secure, scalable backend architecture for multi-region deployment.

Full-Stack Developer
Dec 2020 – Feb 2022
Setel (Petronas) | *Malaysia*

Maintained the loyalty and inventory systems for Malaysia's leading e-wallet.

- Scaled the store ordering system to handle nationwide self-checkout volume.
- Optimized inventory tracking logic to ensure real-time consistency.

OPEN-SOURCE CONTRIBUTIONS

Redis – *Contributor*

- Fixed a critical crash in ZINTER involving SET and ZSET types.
- Optimized integer checks and suggested SDS memory layout improvements.

Rust Coreutils – *Contributor*

- Reimplemented the 'tr' utility to match POSIX/GNU standards, achieving a 100% test pass rate.
- Improved the performance and stability of 'ls' and 'more'.

Godot Engine – *Contributor*

- Resolved memory leaks and UI-related crashes in the core engine.

KEY PROJECTS

radish | High-performance Redis clone in Go.

- Achieved 90% of Redis's performance using a multithreaded architecture.

MIPS Processor | A VHDL implementation of a MIPS processor with forwarding and branch-hazard protection.

- Implemented a full instruction pipeline with hazard mitigation.